

Non-Gaussian Response Distributions

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Overview

GAMs are not limited to Gaussian responses. By specifying a **family** (distribution) and **link function**, we can model count data, binary outcomes, positive continuous data, and more. The general model is:

$$g(\mu_i) = \beta_0 + f_1(x_{1i}) + \cdots + f_p(x_{pi}), \quad y_i \sim \text{Family}(\mu_i, \phi)$$

This vignette demonstrates three common non-Gaussian families:

Family	Link	Response type	Example
Poisson	log	Counts	Species counts, event rates
Binomial	logit	Binary / proportions	Presence-absence, disease status
Gamma	inverse (or log)	Positive continuous	Waiting times, claim sizes

Setup

```
using GAM
using CSV
using StatsAPI: residuals, fitted, r2, deviance
using Statistics: mean
using GLM: LogLink, LogitLink

using DataFrames
using Plots
using Distributions
```

Poisson GAM — count data

Simulating count data

We simulate counts from a Poisson distribution with a smooth log-rate:

$$y_i \sim \text{Poisson}(\mu_i), \quad \log(\mu_i) = 1 + 1.5 \sin(2\pi x_i)$$

```
df_pois = CSV.read("data_poisson.csv", DataFrame)
x = df_pois.x
n = nrow(df_pois)
= 1.0 .+ 1.5 .* sin.(2 .* x)
```

```
300-element Vector{Float64}:
 1.0022515463804877
 1.013013980830874
```

1.014801884538615
 1.0214214724578388
 1.037208399572455
 1.0690982444331225
 1.0742815178198883
 1.165054158814009
 1.1757202903086204
 1.2015521628982468

0.6508219129798084
 0.69797588899615
 0.7278818802184517
 0.7517667995108033
 0.7954284540183503
 0.8368919812276477
 0.8383703243966152
 0.8953919908689525
 0.9675123075811386

Fitting the model

```
m_pois = gam(@gam_formula(y ~ s(x, k = 15, bs = :cr)), df_pois;
              family = Poisson(), link = LogLink())
m_pois
```

Generalized Additive Model

Formula: y ~ 1

Family: Poisson

Link: LogLink

Method: REML

Parametric coefficients:

	Coef.	Std. Error	z	Pr(> z)
(Intercept)	0.939592	0.0468467	20.06	<1e-88

Approximate significance of smooth terms:

Smooth	edf	Ref.df
s(x,bs=cr)	8.00	14

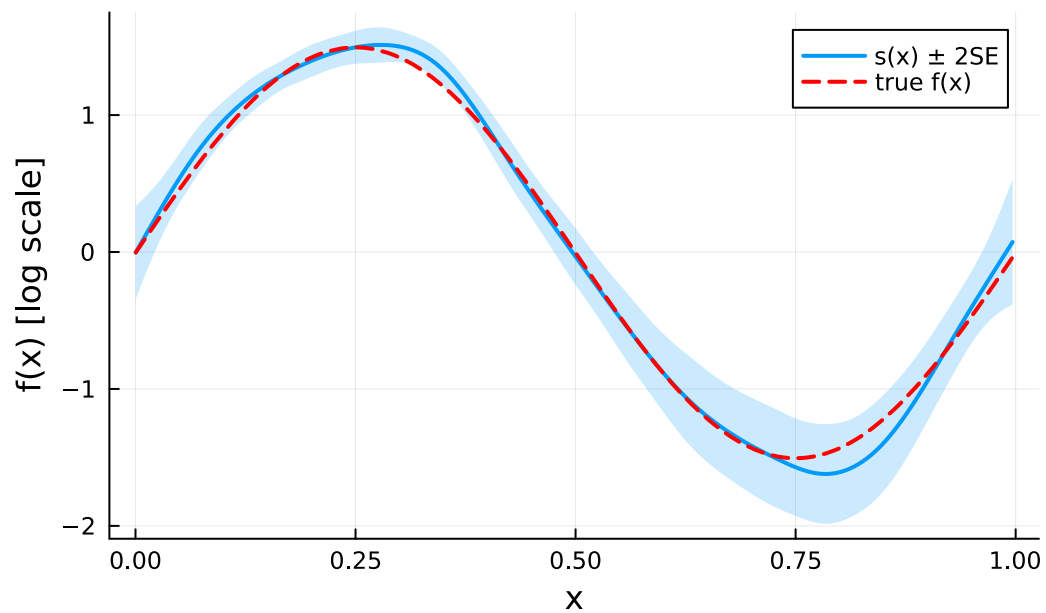
R^2 (adj) = 0.764 Deviance explained = 76.4%
n = 300

Smooth estimate

```
se_pois = smooth_estimates(m_pois; n = 200)
x_grid = se_pois.covariates[:x]

p1 = plot(x_grid, se_pois.estimate;
          ribbon = 2 .* se_pois.se,
          fillalpha = 0.2,
          label = "s(x) ± 2SE",
          linewidth = 2,
          xlabel = "x",
          ylabel = "f(x) [log scale]",
          title = "Poisson GAM - smooth on link scale")
plot!(p1, x,    .- mean( );
      label = "true f(x)",
      linestyle = :dash,
      linewidth = 2,
      color = :red)
p1
```

Poisson GAM — smooth on link scale

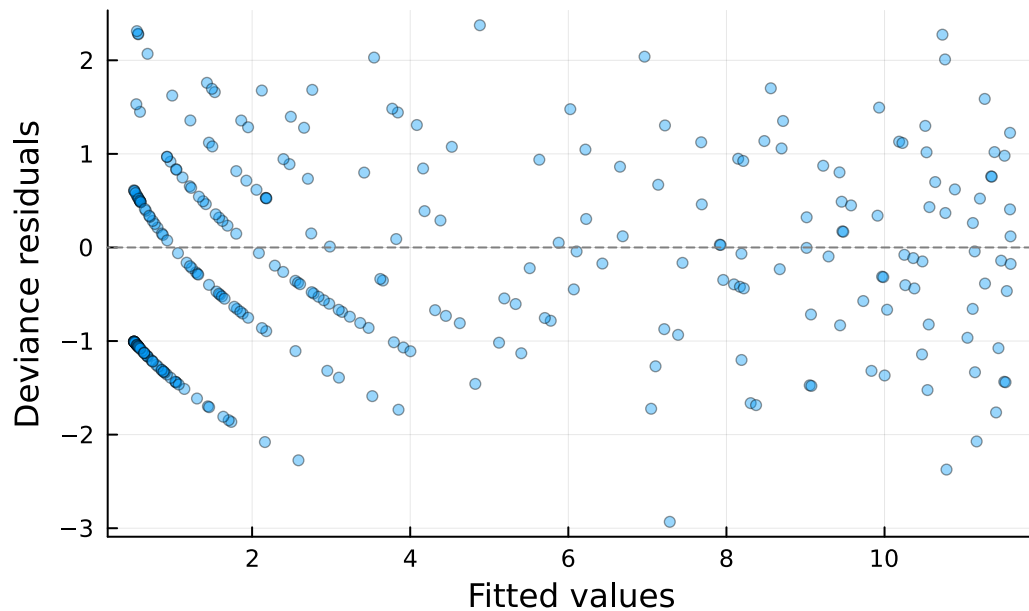


Deviance residuals

```
resid_pois = residuals(m_pois; type = :deviance)

p2 = scatter(fitted(m_pois), resid_pois;
    xlabel = "Fitted values",
    ylabel = "Deviance residuals",
    title = "Poisson GAM - residuals",
    alpha = 0.4,
    markersize = 3,
    label = false)
hline!(p2, [0]; linestyle = :dash, color = :gray, label = false)
p2
```

Poisson GAM — residuals



Binomial GAM — binary data

Simulating binary data

We simulate binary outcomes from a logistic model:

$$y_i \sim \text{Bernoulli}(p_i), \quad \text{logit}(p_i) = -0.5 + 2 \sin(2\pi x_i)$$

```
df_bin = CSV.read("data_binomial.csv", DataFrame)
x_bin = df_bin.x
_bin = -0.5 .+ 2.0 .* sin.(2 .* x_bin)
```

300-element Vector{Float64}:

```
-0.4377380740427925
-0.38854578391368955
-0.373831674500529
-0.3621071915006612
-0.3019464421915505
-0.268943862845824
-0.2102686483145333
-0.15644424465058399
```

-0.15584096852217055
-0.14604202493880453

-0.7494849221391106
-0.7443595705245496
-0.7350585389053363
-0.7139247353867896
-0.7050111267006494
-0.5752349322611175
-0.5557362670383619
-0.5496105296196276
-0.5427940453659735

Fitting the model

```
m_bin = gam(@gam_formula(y ~ s(x, k = 15, bs = :cr)), df_bin;  
            family = Binomial(), link = LogitLink())  
m_bin
```

Generalized Additive Model

Formula: y ~ 1

Family: Binomial

Link: LogitLink

Method: REML

Parametric coefficients:

	Coef.	Std. Error	z	Pr(> z)
(Intercept)	-0.858846	0.159901	-5.37	<1e-07

Approximate significance of smooth terms:

Smooth	edf	Ref.df
s(x,bs=cr)	5.03	14

R^2 (adj) = 0.295 Deviance explained = 25.5%
n = 300

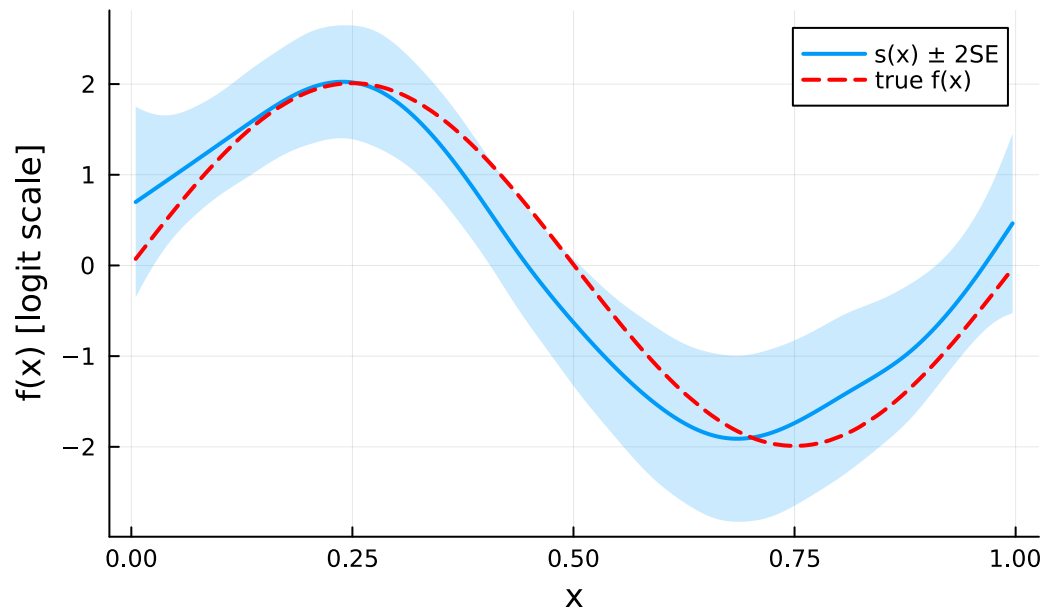
Smooth estimate

```
se_bin = smooth_estimates(m_bin; n = 200)
x_grid_bin = se_bin.covariates[:x]

p3 = plot(x_grid_bin, se_bin.estimate;
          ribbon = 2 .* se_bin.se,
          fillalpha = 0.2,
          label = "s(x) ± 2SE",
          linewidth = 2,
          xlabel = "x",
          ylabel = "f(x) [logit scale]",
          title = "Binomial GAM - smooth on link scale")
plot!(p3, x_bin, _bin .- mean(_bin);
      label = "true f(x)",
      linestyle = :dash,
      linewidth = 2,
      color = :red)

p3
```


Binomial GAM — smooth on link scale

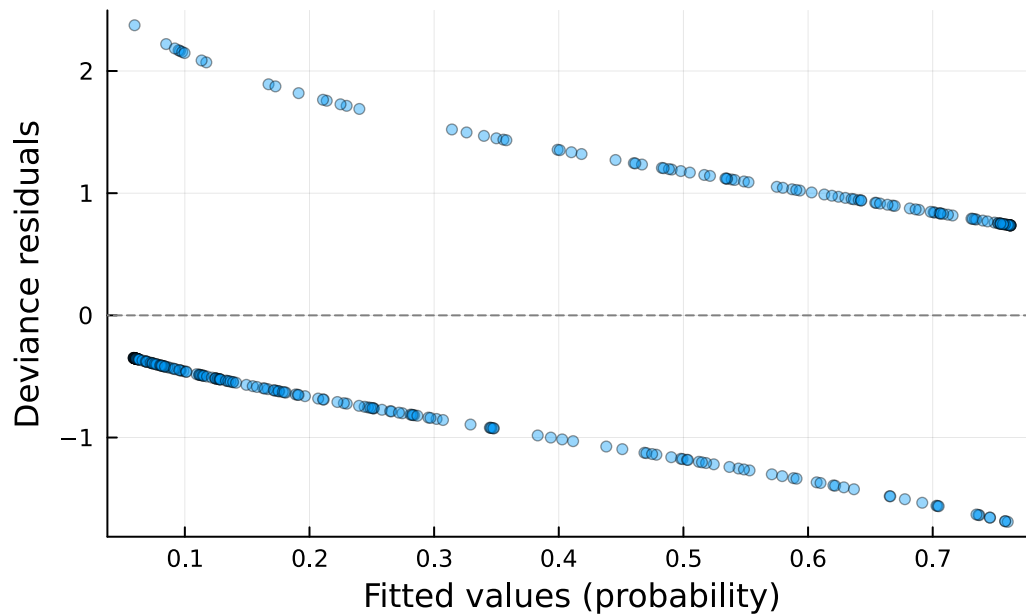


Deviance residuals

```
resid_bin = residuals(m_bin; type = :deviance)

p4 = scatter(fitted(m_bin), resid_bin;
  xlabel = "Fitted values (probability)",
  ylabel = "Deviance residuals",
  title = "Binomial GAM - residuals",
  alpha = 0.4,
  markersize = 3,
  label = false)
hline!(p4, [0]; linestyle = :dash, color = :gray, label = false)
p4
```

Binomial GAM — residuals



Gamma GAM — positive continuous data

Simulating positive continuous data

We simulate positive continuous data from a Gamma distribution with a smooth log-mean:

$$y_i \sim \text{Gamma}(\text{shape}, \text{scale}_i), \quad \log(\mu_i) = 1 + \sin(2\pi x_i)$$

```
df_gam = CSV.read("data_gamma.csv", DataFrame)
x_gam = df_gam.x
_gam = 1.0 .+ sin.(2 .* x_gam)
```

300-element Vector{Float64}:

```
1.000850704315502
1.0039634355393983
1.0264783803611617
1.0383131142818598
1.0751167938723143
1.0766376001001923
1.08050888736706
1.157843246765896
```

```

1.20089524015795
1.2452611220182526

0.8188335469516427
0.819999062363841
0.8356988327736081
0.8772715675927307
0.8829807993731171
0.9580649708004427
0.9693231877198806
0.9749523420975444
0.9923732280689034

```

Fitting the model

We use `Gamma()` with `LogLink()` (log link is more commonly used in practice than the canonical inverse link):

```

m_gam = gam(@gam_formula(y ~ s(x, k = 15, bs = :cr)), df_gam;
           family = Gamma(), link = LogLink())
m_gam

```

Generalized Additive Model

Formula: y ~ 1

Family: Gamma
 Link: LogLink
 Method: REML

Parametric coefficients:

	Coef.	Std. Error	t	Pr(> t)
(Intercept)	1.02611	0.024569	41.76	<1e-99

Approximate significance of smooth terms:

Smooth	edf	Ref.df
--------	-----	--------

s(x,bs=cr)

6.23

14

R^2 (adj) = 0.362 Deviance explained = 41.6%
Scale est. = 0.1811 n = 300

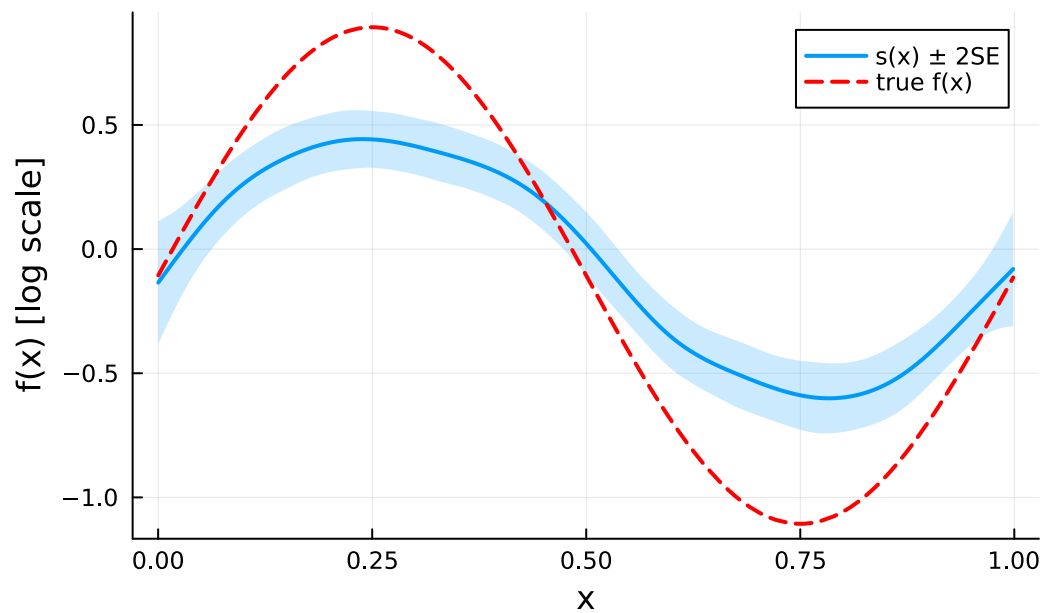
Smooth estimate

```
se_gam = smooth_estimates(m_gam; n = 200)
x_grid_gam = se_gam.covariates[:x]

p5 = plot(x_grid_gam, se_gam.estimate;
          ribbon = 2 .* se_gam.se,
          fillalpha = 0.2,
          label = "s(x) ± 2SE",
          linewidth = 2,
          xlabel = "x",
          ylabel = "f(x) [log scale]",
          title = "Gamma GAM - smooth on link scale")
plot!(p5, x_gam, _gam .- mean(_gam);
      label = "true f(x)",
      linestyle = :dash,
      linewidth = 2,
      color = :red)

p5
```

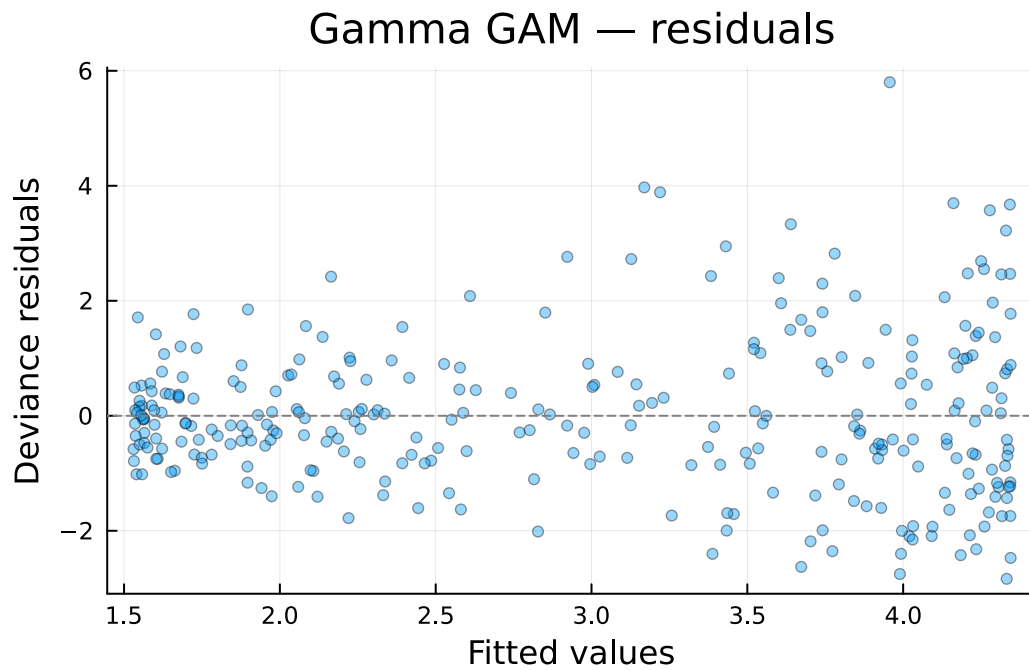
Gamma GAM — smooth on link scale



Deviance residuals

```
resid_gam = residuals(m_gam; type = :deviance)

p6 = scatter(fitted(m_gam), resid_gam;
  xlabel = "Fitted values",
  ylabel = "Deviance residuals",
  title = "Gamma GAM - residuals",
  alpha = 0.4,
  markersize = 3,
  label = false)
hline!(p6, [0]; linestyle = :dash, color = :gray, label = false)
p6
```



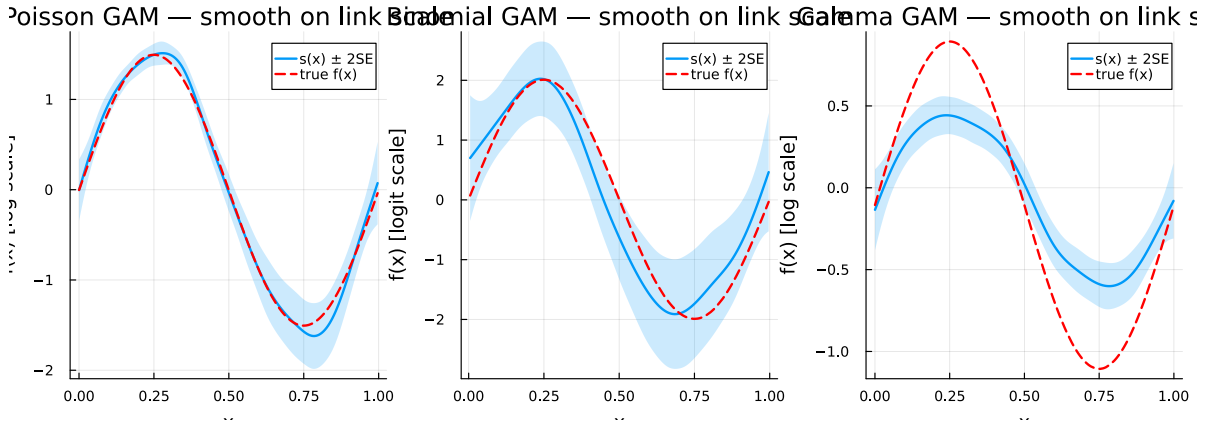
Comparison

```
println("Family      EDF      Dev.Expl(%)  Scale")
println(" " ^ 55)
for (name, m) in [("Poisson", m_pois), ("Binomial", m_bin), ("Gamma", m_gam)]
  e = round(edf(m)[1]; digits = 2)
  de = round(r2(m) * 100; digits = 1)
  sc = round(m.scale; digits = 4)
  println("$($rpad(name, 14))$(lpad(string(e), 6)) $(lpad(string(de), 12)) $(lpad(string(sc), 12))")
end
```

Family	EDF	Dev.Expl(%)	Scale
Poisson	8.0	77.1	1.0
Binomial	5.03	30.7	1.0
Gamma	6.23	37.5	0.1811

All smooth estimates together

```
plot(p1, p3, p5; layout = (1, 3), size = (1000, 350))
```



Summary

In this vignette we:

1. Simulated count data and fitted a **Poisson GAM** with a log link
2. Simulated binary data and fitted a **Binomial GAM** with a logit link
3. Simulated positive continuous data and fitted a **Gamma GAM** with a log link
4. Examined smooth estimates and deviance residuals for each family
5. Compared EDF and deviance explained across families

Each family uses a different link function to map the linear predictor to the mean of the response distribution, but the smooth estimation machinery is the same.